

Name: _____

Materials that you may use during the exam:

1. One page (double sides) cheat sheet (A4). ONLY equations/formulas are allowed.
2. NOT open book
3. NOT open notes (including the printed handouts the instructors distributed in class)
4. NOT open to any other resources (including homework, quizzes, practical exercises).

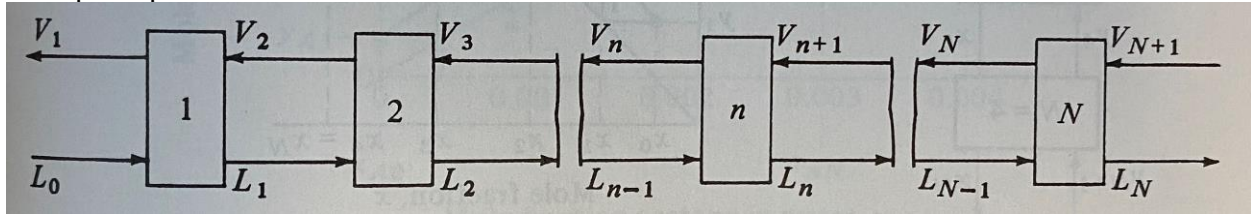
** Violation of the above policy will result in significant grade deduction.*

A. (20 pts) What will be the concentration of oxygen dissolved in water at 298 K when the solution is in equilibrium with air at 1 atm total pressure? The Henry's law constant is 4.38×10^4 atm/mol fraction. (Hint Oxygen composition is 21% in air)

B. (25 pts) A gas mixture at 1 atm pressure containing air and CO_2 is contacted in a single-stage mixer continuously with pure water at 293 K. The two exit gas and liquid streams reach equilibrium. The inlet gas flow rate is 100 kmol/h, with a mole fraction of CO_2 of $y_{A2} = 0.2$. The liquid flow rate entering is 300 kmol water/h. Assume that water does not vaporize to the gas phase. At 293 K, the Henry's law constant is 0.142×10^4 atm/mol fraction. Calculate the amounts and compositions of the two outlet phases.



C. (25 pts) Derive the operation line equation, $(y_{n+1}(x_n))$ for a countercurrent multiple-stage absorption process.



D. (30 pts) Absorption of acetone in a countercurrent stage tower. It is desired to absorb 90% of the acetone in a gas containing 1.0 mol% acetone in air in a countercurrent stage tower. The total inlet gas flow to the tower is 30 kmol/h, and the total inlet pure water flow to be used to absorb the acetone is 90 kmol H₂O/h. the process is to operate isothermally at 300 K and a total pressure of 101.3 kPa. The equilibrium relation for the acetone (A) in the gas-liquid is $y_A = 2.53x_A$. Determine the number of theoretical stages required for this separation. (Hint: use the graphic method and assume the operating line is linear.)

